



MINING DEALER BEST PRACTICE

Alternator, Starting Motor, and ECM Training Test Bench

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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February 2016
1510-4.01-1397

1.0 Introduction

In our workshop, technicians need to be able to verify the assembly and operation of an alternator and starting motor after an overhauling motor. Testing allows the dealer to ensure that the alternator and starting motor will operate properly on the machine. Rotation of an alternator and starting motor provides the most conclusive evaluation.

Most electronic control modules and electronic monitors work with the same strategy. Sometimes it is not possible to find some ECM and monitor failures on the machine. Some customers bring the ECM and monitor to the workshop in order to check and get information.

The alternator, starting motor and ECM testing bench has been designed to test and find faults for reduced cost. The bench also provides opportunities for training.

2.0 Best Practice Description

The alternator, starting motor, and ECM test bench is designed to do the final test of the alternator and starting motor after an overhaul. Also this bench provides testing and flashing of all kind of ECM and monitors for the technician.

The Alternator and starting bench capabilities are as following:

- ❖ Able to measure current and voltage of alternator
- ❖ Able to measure voltage at R terminal
- ❖ Able to measure speed of alternator
- ❖ Able to measure current and voltage of starting motor
- ❖ Able to measure voltage on the solenoid

The ECM test bench capabilities are as following:

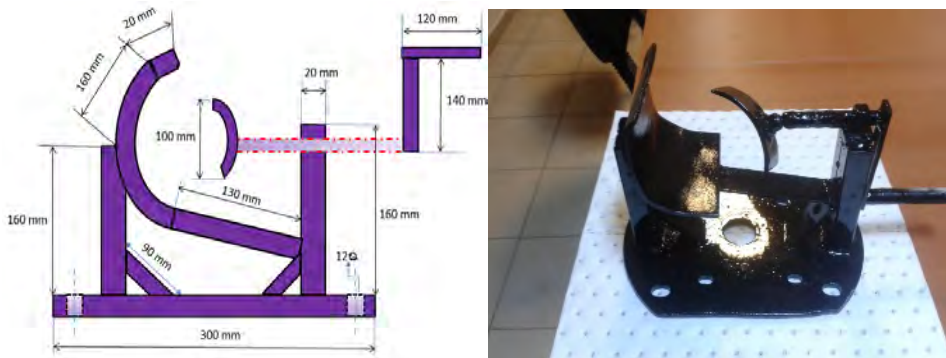
- ❖ Able to connect to all type of ECMs
- ❖ Able to test all type of monitoring systems
- ❖ Able to flash and load configuration to various ECM

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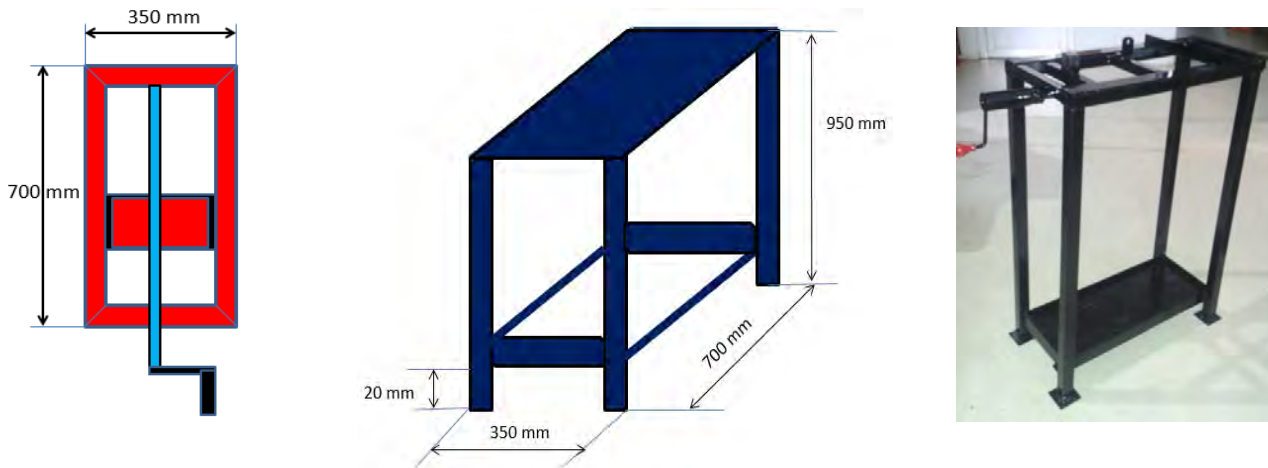
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3.0 Implementation Steps

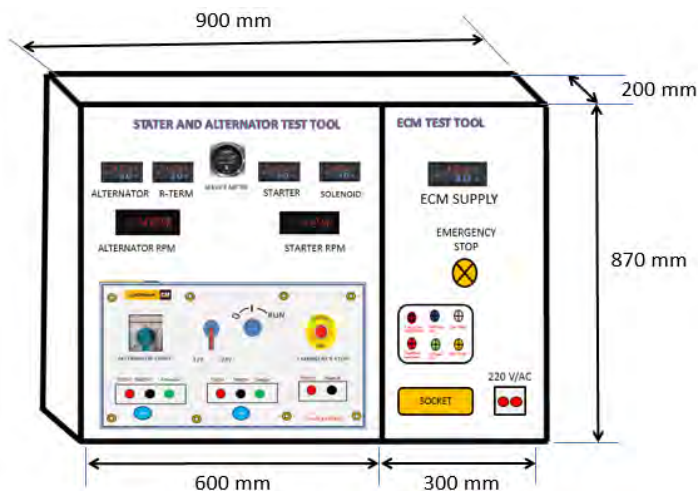
Step-1 Build the holder of starting motor.



Step-2 Build the holder for the alternator and battery box



Step-3 Build the controller of the test bench board



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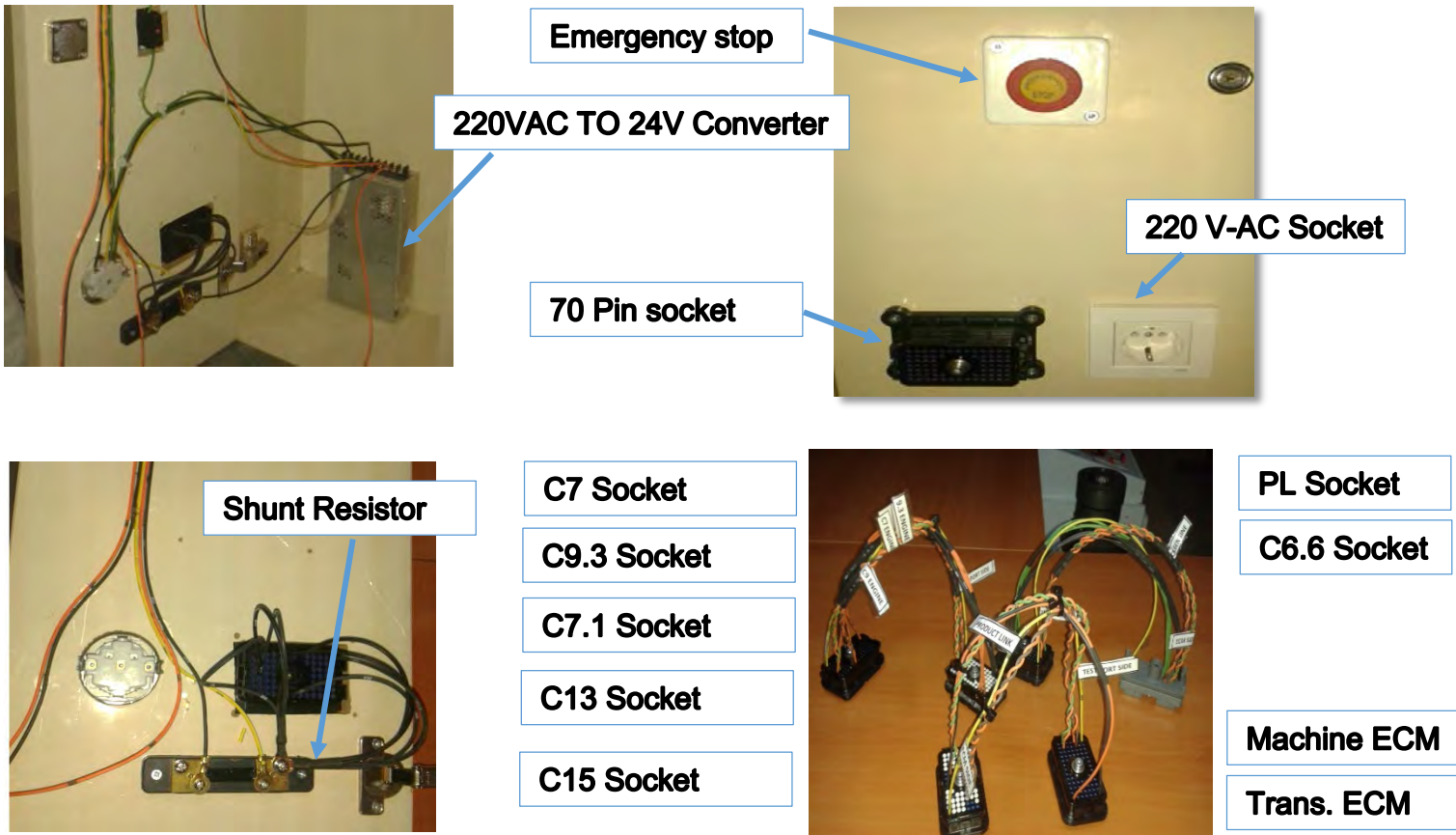
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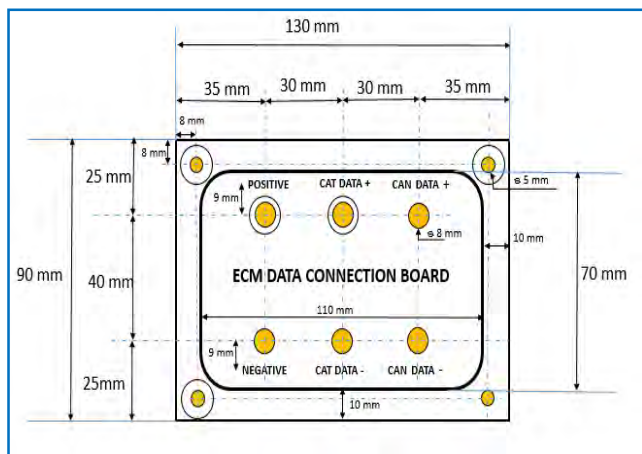
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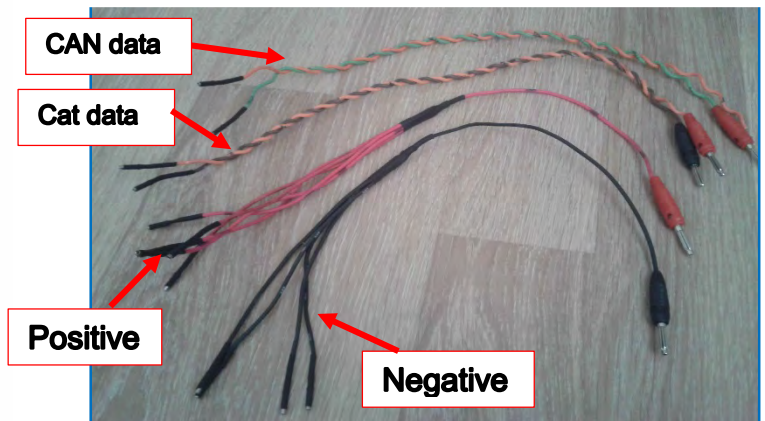
Step-8 Install the wiring for all the devices for the ECM tester according to the electrical schematic



Step-9 Make the ECM connection board for Cat data CAN data and the supply for the ECM



Connection wires for unknown ECM and Monitor

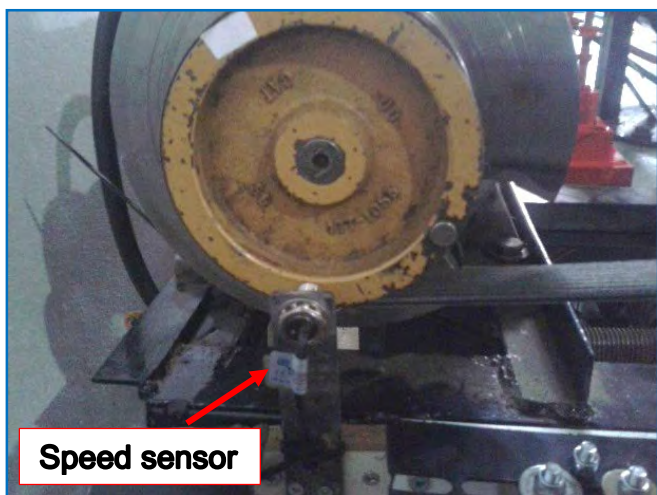
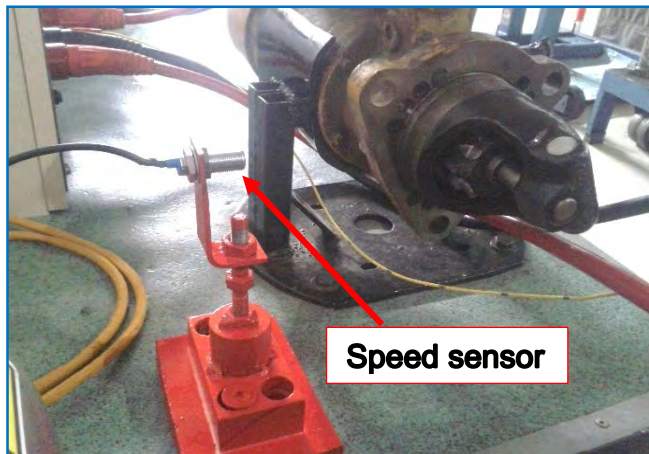


Note: ATA connection wires is are also available inside of the board.

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Step-10 Build the guard protection for the drive system and assembly speed sensor for the alternator and starting motor



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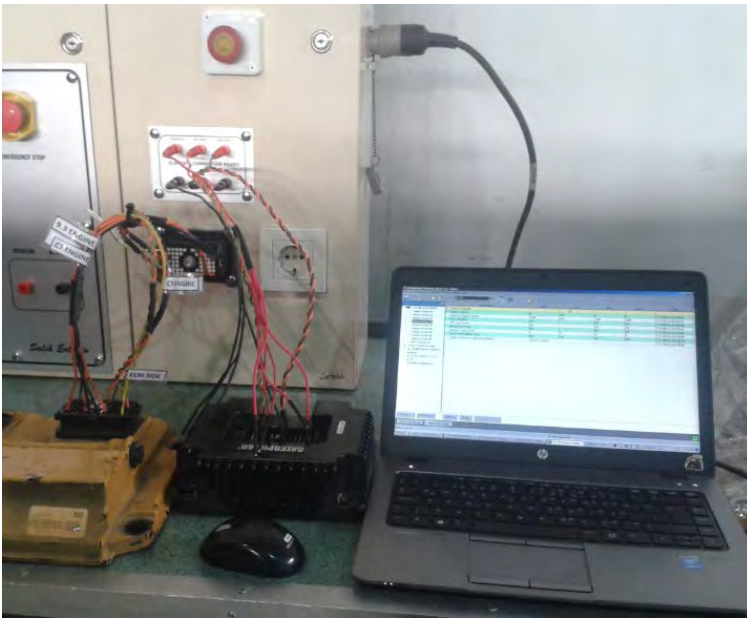
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Step-11 Test the connection between the ECM and Cat ET program



Step-12 Test the alternator and starting motor



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Step-13 Perform the test, file the report before and after overhauling the starting motor and the alternator

ALTERNATOR AND STARTING MOTOR OVERHAUL REPORT			
MACHINE MODEL	330 D	COSTUMER	BORUSAN
SERIAL NUMBER	RAS00198	ENGINE MODEL	C9
ALTERNATOR	NO LOAD	FULL LOAD	RESULT
VOLTAGE	28.4 DC VOLT	23 DC VOLT	OK
CURRENT	13 AMPER	50 A	OK
RPM	1500 RPM	1470 RPM	OK
R-TERMINAL	13.1 DC VOLT	13.8 DC VOLT	OK
MECHANIC PROBLEM	OK	OK	OK
STARTING MOTOR	NO LOAD	FULL LOAD	RESULT
VOLTAGE	28.4 DC VOLT	21 DC VOLT	OK
CURRENT	0	121 AMPER	OK
RPM	0 RPM	5000 RPM	OK
MECHANIC PROBLEM	GEAR CRACK	GEAR NOT FORWARD	GEAR NOT WORKING
SELENOID	NO LOAD	FULL LOAD	RESULT

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VOLTAGE	28.4 VOLT	21 DC VOLT	OK
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4.0 Benefits



The following alternator and starting motor tests can be performed on this bench.

- 1) Perform an alternator output test 24V/12V on the bench
- 2) Perform a regulator test on the bench
- 3) Perform starter test on the bench
- 4) Perform no-load starter test on the bench

The following monitoring and ECM tests can be performed on this bench.

- 1) Test monitoring system input components
- 2) Test monitoring system output components
- 3) Test monitoring system display panel
- 4) Test monitoring system input processor
- 5) Test ECM for input and output component
- 6) Use Cat ET to perform flash programming
- 7) Use Cat ET to perform factory password
- 8) Use Cat ET to perform copy configuration and replace

5.0 Resources Required

Item		Cost (USD)
Material		\$1690
Cat Parts		\$1459.31
Labor	Design	\$250
	Assembly	\$1250
Total		\$4649.31

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6.0 Supporting Attachments / References

- Electrical Schematic
- Technical Drawing
- Presentation of Test Bench

7.0 Related Best Practices

None

8.0 Acknowledgements

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